MATH3701 Statistical Modelling
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Supplementary Proof: Optimality of Natural Splines

This note proves that among all sufficiently smooth functions interpolating prescribed values at a set of knots, the natural cubic spline has the smallest roughness. This result underpins Chapter 11 of the notes (Propositions 11.1 and 11.2).

Setup and statement

Let $t_1 < t_2 < \cdots < t_n$ be n distinct knots and let $y_1, \ldots, y_n$ be prescribed values. We consider functions f in the Sobolev space $W^{2,2}(\mathbb{R})$: functions that are twice differentiable with square-integrable second derivative. The *roughness* of f is

$$J(f) = \int_{-\infty}^{\infty} [f''(t)]^2 \, dt.$$

Theorem 1 (Optimality of the natural cubic spline)**.** *Let f^* be the natural cubic spline with knots at $t_1, \ldots, t_n$ satisfying $f^*(t_i) = y_i$ for $i = 1, \ldots, n$. Then, for any other function $g \in W^{2,2}(\mathbb{R})$ satisfying $g(t_i) = y_i$,*

$$J(g) \geq J(f^*),$$

with equality if and only if $g = f^*$.

Key property of the natural cubic spline

The proof rests on the following orthogonality lemma.

Lemma 1. *Let f^* be the natural cubic spline with $f^*(t_i) = y_i$. For any $h \in W^{2,2}(\mathbb{R})$ with $h(t_i) = 0$ for all i,*

$$\int_{-\infty}^{\infty} (f^*)''(t)\, h''(t)\, dt = 0.$$

Proof. Since f^* is a natural cubic spline, it is a piecewise cubic polynomial with the natural boundary conditions $(f^*)''(t) \to 0$ as $t \to \pm\infty$ (in fact $(f^*)'' = 0$ outside $[t_1, t_n]$). Integrate by parts twice:

$$\int_{-\infty}^{\infty} (f^*)''(t)\, h''(t)\, dt = \big[(f^*)''(t)\, h'(t)\big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} (f^*)'''(t)\, h'(t)\, dt.$$

The boundary term vanishes because $(f^*)'' = 0$ outside the knot range, so $(f^*)''(\pm\infty) = 0$. Integrate by parts again:

$$-\int_{-\infty}^{\infty} (f^*)'''(t)\, h'(t)\, dt = -\big[(f^*)'''(t)\, h(t)\big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} (f^*)''''(t)\, h(t)\, dt.$$

Since f^* is linear outside the knot range, $(f^*)''' = 0$ there, so the boundary term vanishes.

Inside the knot range, a cubic spline has $(f^*)''''(t) = 0$ on each open interval (t_{i-1}, t_i). Thus the integral reduces to a sum of contributions at the knots. Specifically, if we write the fourth derivative in the distributional sense, the jumps in $(f^*)'''$ at each knot t_i contribute a Dirac mass:

$$\int_{-\infty}^{\infty} (f^*)''''(t)\, h(t)\, dt = \sum_{i=1}^{n} \big[(f^*)'''(t_i^+) - (f^*)'''(t_i^-)\big] h(t_i) = 0,$$

since $h(t_i) = 0$ for all i. Hence $\int (f^*)'' h''\, dt = 0$. □

Proof of Theorem 1

Let g be any function in $W^{2,2}(\mathbb{R})$ with $g(t_i) = y_i$, and set $h = g - f^*$. Then $h(t_i) = 0$ for all i. We decompose:

$$J(g) = \int_{-\infty}^{\infty} [g''(t)]^2\,dt = \int_{-\infty}^{\infty} \big[(f^*)''(t) + h''(t)\big]^2\,dt$$
$$= J(f^*) + 2\int_{-\infty}^{\infty} (f^*)''(t)\,h''(t)\,dt + J(h).$$

By Lemma 1, the cross-term vanishes, giving

$$J(g) = J(f^*) + J(h) \geq J(f^*),$$

since $J(h) = \int [h'']^2\,dt \geq 0$.

Equality holds if and only if $J(h) = 0$, i.e., $h'' = 0$ everywhere, meaning h is a linear function. Since h is linear and vanishes at the $n \geq 2$ distinct knots, $h \equiv 0$. Hence $g = f^*$, proving uniqueness. □

Proof of Proposition 11.1: representations of natural splines

Throughout, $t_1 < \cdots < t_n$ are the knots (written $t_1, \ldots, t_m$ in the notes) with $n \geq 2$. Recall that a *natural cubic spline* is a piecewise cubic function with knots t_i whose first and second derivatives are continuous, and for which $f'' = f''' = 0$ on $(-\infty, t_1]$ and $[t_n, \infty)$ (so f is linear outside the knots). A *natural linear spline* is a continuous piecewise linear function with $f' = 0$ on $(-\infty, t_1]$ and $[t_n, \infty)$ (so f is constant outside the knots). We prove that these are *exactly* the functions of the stated forms.

Cubic case

Claim. A function f is a natural cubic spline with knots $t_1, \ldots, t_n$ if and only if

$$f(t) = a_0 + a_1 t + \sum_{i=1}^{n} b_i |t - t_i|^3 \quad \text{with} \quad \sum_{i=1}^{n} b_i = 0, \quad \sum_{i=1}^{n} b_i t_i = 0.$$

Proof. We use two elementary facts. First, since $(x)_+^3 = \frac{1}{2}\big(x^3 + |x|^3\big)$,

$$\sum_i d_i (t - t_i)_+^3 = \tfrac{1}{2}\sum_i d_i (t - t_i)^3 + \tfrac{1}{2}\sum_i d_i |t - t_i|^3. \tag{1}$$

Second, expanding the cube,

$$\sum_i d_i (t - t_i)^3 = t^3 \sum_i d_i - 3t^2 \sum_i d_i t_i + 3t \sum_i d_i t_i^2 - \sum_i d_i t_i^3, \tag{2}$$

which is a polynomial of degree at most 1 in t whenever $\sum_i d_i = \sum_i d_i t_i = 0$.

($\Rightarrow$) Let f be a natural cubic spline. Let $d_i = \frac{1}{6}\big[f'''(t_i^+) - f'''(t_i^-)\big]$ be one sixth of the jump in the third derivative at t_i. Then $g(t) = f(t) - \sum_i d_i (t - t_i)_+^3$ is piecewise cubic with no jump in g, g', g'', g''' at any knot, so g is a single cubic polynomial and

$$f(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3 + \sum_{i=1}^{n} d_i (t - t_i)_+^3.$$

For $t \le t_1$ every truncated term vanishes, so $f'' = 2c_2 + 6c_3 t = 0$ on $(-\infty, t_1]$, which forces $c_2 = c_3 = 0$. For $t \ge t_n$ all truncated terms equal $(t - t_i)^3$, so

$$f'''(t) = 6\sum_i d_i = 0, \qquad f''(t) = 6\sum_i d_i(t - t_i) = 6t\sum_i d_i - 6\sum_i d_i t_i = 0$$

for all $t \ge t_n$. Hence $\sum_i d_i = 0$ and $\sum_i d_i t_i = 0$. By (1) and (2),

$$f(t) = c_0 + c_1 t + \tfrac{1}{2}\big(\alpha + \beta t\big) + \tfrac{1}{2}\sum_i d_i |t - t_i|^3,$$

where $\alpha = -\sum_i d_i t_i^3$ and $\beta = 3\sum_i d_i t_i^2$ are constants. This is the stated form with $b_i = d_i/2$, $a_0 = c_0 + \alpha/2$, $a_1 = c_1 + \beta/2$, and $\sum_i b_i = \sum_i b_i t_i = 0$.

($\Leftarrow$) Let f be of the stated form. Each $|t - t_i|^3$ is a cubic polynomial on each side of t_i and has continuous first and second derivatives (its second derivative is $6|t - t_i|$), so f is a cubic spline with knots t_i. For $t \ge t_n$ we have $|t - t_i|^3 = (t - t_i)^3$, and for $t \le t_1$ we have $|t - t_i|^3 = -(t - t_i)^3$. In both cases (2) together with $\sum b_i = \sum b_i t_i = 0$ shows that $\sum_i b_i |t - t_i|^3$ is a polynomial of degree at most 1, so f is linear there and $f'' = f''' = 0$. Thus f is a natural cubic spline. □

Linear case

Claim. A function f is a natural linear spline with knots $t_1, \dots, t_n$ if and only if

$$f(t) = a_0 + \sum_{i=1}^{n} b_i |t - t_i| \quad \text{with} \quad \sum_{i=1}^{n} b_i = 0.$$

Proof. The argument is the same, using $(x)_+ = \frac{1}{2}(x + |x|)$.

($\Rightarrow$) A continuous piecewise linear f can be written $f(t) = c_0 + c_1 t + \sum_i d_i (t - t_i)_+$, where d_i is the jump in f' at t_i. For $t \le t_1$, $f' = c_1 = 0$. For $t \ge t_n$, $f' = \sum_i d_i = 0$. Then

$$\sum_i d_i (t - t_i)_+ = \tfrac{1}{2}\sum_i d_i (t - t_i) + \tfrac{1}{2}\sum_i d_i |t - t_i| = -\tfrac{1}{2}\sum_i d_i t_i + \tfrac{1}{2}\sum_i d_i |t - t_i|,$$

so $f(t) = a_0 + \sum_i b_i |t - t_i|$ with $b_i = d_i/2$, $a_0 = c_0 - \frac{1}{2}\sum_i d_i t_i$ and $\sum_i b_i = 0$.

($\Leftarrow$) The function f is continuous and piecewise linear with knots t_i. For $t \ge t_n$, $\sum_i b_i |t - t_i| = \sum_i b_i (t - t_i) = -\sum_i b_i t_i$, and for $t \le t_1$ it equals $\sum_i b_i (t_i - t) = \sum_i b_i t_i$; both are constant, so $f' = 0$ outside $[t_1, t_n]$. □

Parameter count and interpolation

In the cubic case the representation has $n + 2$ parameters $(a_0, a_1, b_1, \dots, b_n)$ subject to 2 linear constraints, leaving n free parameters (and $n + 1 - 1 = n$ in the linear case), in agreement with the degrees of freedom $m = n$ of a natural spline in the notes. The interpolation conditions $f(t_i) = y_i$ are n further linear equations, and they have a unique solution: if two natural splines interpolate the same data, their difference h is a natural spline with $h(t_i) = 0$ for all i, and applying the Lemma above with $f^* = h$ gives $\int (h'')^2\, dt = 0$ (the Lemma used only that its first argument is a natural cubic spline), so h is linear and vanishing at two or more knots, i.e. $h \equiv 0$. The square linear system for the free parameters therefore has a trivial null space, so it is invertible and a natural spline interpolant exists for every $\mathbf{y}$.